

AUA2

# Graphing, Proportional Reasoning and Equations/Inequalities

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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## How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# Items

1.

A city planner who has worked downtown for ten years marks a new bus stop at  $(-3, 4)$  on a grid map where each unit is one block. In which quadrant is the stop?

- A. Quadrant IV
- B. Quadrant III
- C. Quadrant I
- D. Quadrant II

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2.

A hiking trail that is popular with families on weekends climbs from 300 feet elevation at mile 2 to 1,100 feet at mile 6. What is the average slope in feet per mile?

- A. 200 feet per mile
- B. About 183 feet per mile
- C. 800 feet per mile
- D. 0.005 feet per mile

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3. [TRANSFER — reversed stem / different variable]

A line has slope  $-3$  and passes through  $(1, 5)$ . What is  $y$  when  $x = 4$ ?

- A.  $-7$
- B.  $2$
- C.  $14$
- D.  $-4$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

4.

A gym that has three locations and opened in 2015 charges a \$50 sign-up fee plus \$25 per month. Which equation gives the total cost  $y$  after  $x$  months?

- A.  $y = 25x + 50$
- B.  $y = 25x$
- C.  $y = 50x + 25$
- D.  $y = 75x$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

5.

A café that roasts its own beans tracks cups of drip coffee sold ( $x$ ) and revenue in dollars ( $y$ ). Which set of data shows a proportional relationship?

- A. (2, 4), (5, 25), (8, 64)
- B. (2, 6), (5, 15), (8, 24)
- C. Any set where  $y$  goes up whenever  $x$  goes up.
- D. (2, 7), (5, 16), (8, 25)

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

6. [TRANSFER — reversed stem / different variable]

$y$  is proportional to  $x$ , and  $y = 18$  when  $x = 4$ . What is  $y$  when  $x = 10$ ?

- A. 72
- B. 24
- C. 45
- D. About 2.2

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

7.

A grocery store with a loyalty program sells a 12-ounce bag of rice for \$3.00 and a 20-ounce bag for \$4.40. Loyalty members also get free parking. Which is the better buy?

- A. The 12-ounce bag, by \$0.03 per ounce.
- B. The 20-ounce bag, saving \$1.40.
- C. The 20-ounce bag, by \$0.03 per ounce.
- D. The 12-ounce bag, because its total price is lower.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

8.

An architect's model, built for a client in the tech industry, uses a scale of 2 inches = 5 feet. A wall on the model is 14 inches long. How long is the real wall?

- A. 5.6 feet
- B. 35 feet
- C. 70 feet
- D. 17 feet

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

9.

At a theater that seats 400, three friends each paid the ticket price minus a \$4 coupon and spent \$21 in total. Solve  $3(x - 4) = 21$  for the ticket price  $x$ .

- A. \$7
- B. \$11
- C.  $\$25/3$  (about \$8.33)
- D. \$3

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

10. [TRANSFER — reversed stem / different variable]

The equation  $5x + b = 2$  has solution  $x = -2$ . What is  $b$ ?

- A. 12
- B. 2
- C. -8
- D. -12

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

11.

A food-truck owner whose truck is painted bright yellow models her remaining budget with  $200 - 15x \geq 50$ , where  $x$  is the number of permit hours she buys. Which values of  $x$  work?

- A.  $x \leq$  about 16.7
- B.  $x \geq 10$
- C.  $x \leq 10$
- D.  $x \geq -10$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

12. [TRANSFER — reversed stem / different variable]

Which step, applied to both sides of an inequality, requires reversing the inequality sign?

- A. Any step that uses a negative number, including adding  $-4$ .
- B. Multiplying both sides by 3.
- C. Subtracting 5 from both sides.
- D. Dividing both sides by  $-2$ .

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer D Concept 1

A city planner ~~who has worked downtown for ten years~~ marks a new bus stop at  $(-3, 4)$  on a grid map ~~where each unit is one block~~. In which quadrant is the stop?

**Target word:** quadrant **Decides:** x negative, y positive.

|   |                    |                                                 |
|---|--------------------|-------------------------------------------------|
| A | Inverted operation | Reads the pair as $(4, -3)$ , swapping x and y. |
| B | Confusion set      | Quadrant III needs both coordinates negative.   |
| C | Dropped element    | Drops the negative sign on x.                   |
| D | <b>Correct</b>     | $(-, +)$ is quadrant II.                        |

## 2. Answer A Concept 2

A hiking trail ~~that is popular with families on weekends~~ climbs from 300 feet elevation at mile 2 to 1,100 feet at mile 6. What is the average slope in feet per mile?

**Target word:** rate of climb **Decides:**  $(1,100 - 300) / (6 - 2)$ .

|   |                    |                                                                   |
|---|--------------------|-------------------------------------------------------------------|
| A | <b>Correct</b>     | $800 / 4 = 200$ .                                                 |
| B | Wrong stem number  | Divides one point's values, $1,100 / 6$ , instead of the changes. |
| C | Dropped element    | Uses the rise but drops the run.                                  |
| D | Inverted operation | Divides run by rise: $4 / 800$ .                                  |

## 3. Answer D Concept 2 · transfer

A line has slope  $-3$  and passes through  $(1, 5)$ . What is y when  $x = 4$ ?

**Target word:** solve for y using slope **Decides:** Run of 3 gives a change of  $-9$ .

|   |                    |                                                         |
|---|--------------------|---------------------------------------------------------|
| A | Wrong stem number  | Uses $x = 4$ as the run instead of $4 - 1$ .            |
| B | Dropped element    | Applies the slope for one unit of run instead of three. |
| C | Inverted operation | Adds 3 per unit instead of subtracting.                 |
| D | <b>Correct</b>     | $5 + (-3)(3) = -4$ .                                    |

## 4. Answer A Concept 3

A gym ~~that has three locations and opened in 2015~~ charges a \$50 sign-up fee plus \$25 per month. Which equation gives the total cost y after x months?

**Target word:** model the cost **Decides:** \$25 is the rate; \$50 is the starting value.

|   |                    |                                             |
|---|--------------------|---------------------------------------------|
| A | <b>Correct</b>     | Rate times months plus the one-time fee.    |
| B | Dropped element    | Drops the sign-up fee.                      |
| C | Inverted operation | Swaps the rate and the starting value.      |
| D | Adjacent category  | Adds fee and rate into a proportional rate. |

## 5. Answer B Concept 4

A café that roasts its own beans tracks cups of drip coffee sold ( $x$ ) and revenue in dollars ( $y$ ). Which set of data shows a proportional relationship?

**Target word:** identify proportionality **Decides:** Constant  $y/x$  ratio through the origin.

|          |                   |                                                                     |
|----------|-------------------|---------------------------------------------------------------------|
| A        | Confusion set     | $y = x^2$ , not a constant ratio.                                   |
| <b>B</b> | <b>Correct</b>    | $y/x = 3$ every time; $y = 3x$ .                                    |
| C        | Absolute language | Increasing is not the same as a constant ratio.                     |
| D        | Adjacent category | Linear ( $y = 3x + 1$ ) but not proportional; it misses the origin. |

## 6. Answer C Concept 4 • transfer

$y$  is proportional to  $x$ , and  $y = 18$  when  $x = 4$ . What is  $y$  when  $x = 10$ ?

**Target word:** solve for  $y$  **Decides:**  $k = 18 / 4 = 4.5$ .

|          |                    |                                                   |
|----------|--------------------|---------------------------------------------------|
| A        | Wrong stem number  | Multiplies the given pair, $18 \times 4$ .        |
| B        | Adjacent category  | Additive reasoning: adds 6 because $x$ went up 6. |
| <b>C</b> | <b>Correct</b>     | $4.5 \times 10 = 45$ .                            |
| D        | Inverted operation | Inverts the ratio: $10 \times 4 / 18$ .           |

## 7. Answer C Concept 5

A grocery store with a loyalty program sells a 12-ounce bag of rice for \$3.00 and a 20-ounce bag for \$4.40. Loyalty members also get free parking. Which is the better buy?

**Target word:** lower cost per ounce **Decides:**  $\$3.00/12 = \$0.25$ ;  $\$4.40/20 = \$0.22$ .

|          |                    |                                                    |
|----------|--------------------|----------------------------------------------------|
| A        | Inverted operation | Right difference, reversed direction.              |
| B        | Wrong stem number  | \$1.40 is the gap in total price, not a saving.    |
| <b>C</b> | <b>Correct</b>     | $\$0.22 < \$0.25$ per ounce.                       |
| D        | Dropped element    | Drops the quantity; total price is not unit price. |

## 8. Answer B Concept 6

An architect's model, built for a client in the tech industry, uses a scale of 2 inches = 5 feet. A wall on the model is 14 inches long. How long is the real wall?

**Target word:** actual length **Decides:**  $2/5 = 14/x$ .

|          |                    |                                                          |
|----------|--------------------|----------------------------------------------------------|
| A        | Inverted operation | Sets the ratio upside down: $14 \times 2 / 5$ .          |
| <b>B</b> | <b>Correct</b>     | $14 / 2 = 7$ ; $7 \times 5 = 35$ .                       |
| C        | Dropped element    | Multiplies $14 \times 5$ and drops the 2 inches.         |
| D        | Adjacent category  | Additive reasoning: adds the 3-unit gap between 2 and 5. |

### 9. Answer B Concept 7

~~At a theater that seats 400,~~ three friends each paid the ticket price minus a \$4 coupon and spent \$21 in total. Solve  $3(x - 4) = 21$  for the ticket price  $x$ .

**Target word:** solve the equation **Decides:** Divide by 3, then add 4.

|          |                    |                                                |
|----------|--------------------|------------------------------------------------|
| A        | Dropped element    | Stops at $x - 4 = 7$ .                         |
| <b>B</b> | <b>Correct</b>     | $x - 4 = 7$ , so $x = 11$ .                    |
| C        | Confusion set      | Distributes the 3 to $x$ only: $3x - 4 = 21$ . |
| D        | Inverted operation | Subtracts 4 instead of adding it back.         |

### 10. Answer A Concept 7 • transfer

The equation  $5x + b = 2$  has solution  $x = -2$ . What is  $b$ ?

**Target word:** solve for the constant **Decides:**  $5(-2) + b = 2$ .

|          |                    |                                    |
|----------|--------------------|------------------------------------|
| <b>A</b> | <b>Correct</b>     | $-10 + b = 2$ , so $b = 12$ .      |
| B        | Wrong stem number  | The right-hand side, not $b$ .     |
| C        | Inverted operation | Uses $+10$ for $5(-2)$ .           |
| D        | Dropped element    | Drops the sign when moving $-10$ . |

### 11. Answer C Concept 8

~~A food-truck owner whose truck is painted bright yellow~~ models her remaining budget with  $200 - 15x \geq 50$ , where  $x$  is the number of permit hours she buys. Which values of  $x$  work?

**Target word:** solution set **Decides:** Dividing by  $-15$  flips the sign.

|          |                    |                                                            |
|----------|--------------------|------------------------------------------------------------|
| A        | Inverted operation | Adds 50 instead of subtracting: $250 / 15$ .               |
| B        | Dropped element    | Forgets to flip the sign.                                  |
| <b>C</b> | <b>Correct</b>     | $-15x \geq -150$ , divide by $-15$ and flip: $x \leq 10$ . |
| D        | Confusion set      | Flips the sign of the number instead of the inequality.    |

### 12. Answer D Concept 8 • transfer

Which step, applied to both sides of an inequality, requires reversing the inequality sign?

**Target word:** when to flip **Decides:** Multiplying or dividing by a negative.

|          |                   |                                                      |
|----------|-------------------|------------------------------------------------------|
| A        | Absolute language | Only multiplying or dividing by a negative flips it. |
| B        | Adjacent category | A positive multiplier keeps the sign.                |
| C        | Confusion set     | Adding or subtracting never flips the sign.          |
| <b>D</b> | <b>Correct</b>    | Division by a negative reverses order.               |

## Triage grid

Write each item number in the box it belongs to.

|                       | <b>Right</b>                                          | <b>Wrong</b>                                                                          |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owled.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| <b>Trap</b>        | <b>What it does</b>                                          | <b>In this drill</b> | <b>Your misses</b> |
|--------------------|--------------------------------------------------------------|----------------------|--------------------|
| Dropped element    | Leaves out a required term or step.                          | 9                    |                    |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 5                    |                    |
| Adjacent category  | The right idea in the wrong bucket.                          | 5                    |                    |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 10                   |                    |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 2                    |                    |
| Confusion set      | Every option is real; the test is exact role-matching.       | 5                    |                    |
| Camouflage         | Answers a stem fact that was never the question.             | 0                    |                    |

# Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                               | Rule                                                                                                                   | Trap                                                                | Tell                                                         |
|---|---------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------|
| 1 | <b>Coordinate pairs and quadrants</b> | An ordered pair is (x, y): horizontal first, vertical second; signs place it — I (+,+), II (-,+), III (-,-), IV (+,-). | Reading y first or ignoring a sign.                                 | A point with at least one negative coordinate.               |
| 2 | <b>Slope as rate of change</b>        | Slope = change in y / change in x between two points.                                                                  | Dividing run by rise, or dividing one y-value by one x-value.       | Two points or a start and end reading.                       |
| 3 | <b>Slope-intercept form</b>           | In $y = mx + b$ , m is the rate per unit and b is the starting value when $x = 0$ .                                    | Swapping the fixed fee and the rate.                                | A one-time fee plus a per-unit charge.                       |
| 4 | <b>Proportional relationships</b>     | A relationship is proportional only if $y/x$ is constant, so $y = kx$ and the graph passes through the origin.         | Calling any straight line, or any increasing pattern, proportional. | "Is it proportional?" or "y varies directly with x."         |
| 5 | <b>Unit rate comparison</b>           | Compare offers by cost per one unit (price ÷ quantity).                                                                | Choosing the lower total price.                                     | Two sizes at two prices.                                     |
| 6 | <b>Scale and proportions</b>          | Set up equal ratios in the same order (model/actual = model/actual) and solve.                                         | Adding the difference instead of scaling (additive reasoning).      | A scale such as 2 in = 5 ft.                                 |
| 7 | <b>Solving linear equations</b>       | Undo operations in reverse order, applying the same step to both sides; distribute before combining.                   | Distributing to only the first term or using the wrong inverse.     | A multi-step equation or a word problem with a grouped cost. |
| 8 | <b>Inequality sign reversal</b>       | Multiplying or dividing both sides of an inequality by a negative number reverses the sign.                            | Forgetting to flip, or flipping for adding a negative.              | A negative coefficient on the variable.                      |

**Course scope (public description):** Graphing, Proportional Reasoning and Equations/Inequalities is a performance-based assessment that evaluates a student's portfolio of work. This portfolio includes the student's responses to various prompts and an original lesson plan for each of the mathematics modules such as coordinate pairs and graphing, ratios and proportional reasoning, and equations and inequalities.